

JECRC UNIVERSITY, JAIPUR

Department of Computer Science & Engineering

PROJECT REPORT

ExamBuddy

A Full-Stack Web Platform for Sharing Study Materials Between Teachers and Students

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology

in Computer Science and Engineering

Submitted By:

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Under the Supervision of:

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1. Introduction

ExamBuddy is a lightweight, full-stack web application developed to streamline the distribution and access of academic study materials within an educational institution. The platform acts as a centralized digital repository that bridges the gap between educators and students, enabling teachers to upload and manage learning resources while students can seamlessly browse and download them.

Built with a modern technology stack — React on the frontend and Node.js with Express on the backend — ExamBuddy is designed to be fast, intuitive, and easy to maintain. It addresses a common challenge in academic environments: the lack of a structured, role-specific portal for sharing course materials, previous year question papers, notes, and other exam-preparatory content.

The application implements a Role-Based Access Control (RBAC) system where Teachers are empowered to upload and manage materials, while Students can view and download resources relevant to their subjects. Each uploaded item is tagged with metadata such as Subject ID, Material Type, Important Topics, and the uploader's name — making the library well-organized and easy to navigate.

ExamBuddy is particularly significant for institutions seeking a simple, deployable internal tool that does not require heavy infrastructure or a commercial subscription to third-party platforms. It promotes organized academic resource-sharing, reduces redundancy, and ensures that all students have equitable access to learning materials.

2. Feasibility Study

2.1 Need

In academic institutions, study materials such as lecture notes, solved question papers, syllabus documents, and reference PDFs are often shared informally through messaging apps or email. This leads to loss of resources, unequal access, and disorganized file management. ExamBuddy provides a structured, reliable, and role-aware alternative.

2.2 Significance

ExamBuddy reduces dependency on informal channels, ensures that all students receive the same quality of materials, and gives teachers a professional interface to manage their uploads. The system also maintains a persistent record of all uploaded materials, which is useful for future academic reference.

2.3 Feasibility Analysis

Technical Feasibility:

The project is built using React (Vite), Node.js, Express, and Multer — all widely adopted, open-source technologies. The lightweight JSON-based database is suitable for an academic prototype and can be upgraded to MongoDB or MySQL for production use.

Operational Feasibility:

The interface is clean and user-friendly. Role-based dashboards for teachers and students ensure that each user type sees only what is relevant to them, minimizing confusion and operational errors.

Economic Feasibility:

As an academic project using entirely free and open-source tools, the cost of development is negligible. Deployment can be achieved on free hosting platforms such as Vercel (frontend) and Render or Railway (backend).

3. Project Objectives and Features

3.1 Objectives

- To develop a centralized, role-based web portal for distributing academic study materials.
- To allow teachers to upload, label, and manage study resources through a structured form interface.
- To enable students to browse, view, and download all available resources for exam preparation.
- To implement secure, role-specific routing that prevents unauthorized access.
- To build a lightweight, deployable system using open-source technologies with no licensing cost.

3.2 Key Features

- Role-Based Access Control (RBAC) with Teacher and Student dashboards; protected routing prevents unauthorized access.
- Teacher Dashboard: Upload materials (PDFs, notes, question papers) with metadata — Title, Subject ID, Material Type, Important Topics, and Uploader Name. Files are stored with a 10 MB per-file size limit.
- Student Dashboard: Browse all uploaded study materials, view metadata, and download files directly for offline exam preparation.
- Persistent material metadata stored via a lightweight JSON database (server_db.json), retained across server restarts.
- RESTful API endpoints: GET /api/materials (fetch all materials) and POST /api/upload (upload new material).
- Concurrent frontend and backend execution using a single npm run dev command via the concurrently package.
- Responsive UI with Lucide React icons for a modern, clean appearance across desktop and mobile browsers.

4. Tools, Technologies, and System Requirements

4.1 Technologies Used

Category	Technology / Tool
Frontend	React.js (Vite), React Router DOM, Lucide React
Backend	Node.js, Express.js
File Handling	Multer (multipart/form-data, 10 MB limit, /uploads directory)
Database	JSON File (server_db.json) — lightweight local metadata storage
Dev Tools	concurrently, npm, Git & GitHub
IDE	Visual Studio Code (VS Code)

Table 4.1: Tools and Technologies Used

4.2 Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 256 GB or more
- Internet Connection: Required for npm package installation and deployment

4.3 Software Requirements

- Operating System: Windows 10/11, macOS, or Linux
- Node.js v18 or above and npm v9 or above
- Browser: Latest version of Chrome, Firefox, or Edge

4.4 Functional Requirements

- Users must be able to select a role (Teacher or Student) and access role-specific dashboards.
- Teachers must be able to upload files along with metadata through a structured form interface.
- Students must be able to view all uploaded materials and download them directly.
- The system must store uploaded file metadata persistently across server sessions.
- The backend API must correctly serve material listings and handle file uploads.

4.5 Non-Functional Requirements

- The application should be responsive and accessible on both desktop and mobile browsers.
- File uploads must be validated for type and size (maximum 10 MB per file).

- Page load and API response times should remain under 3 seconds.
- Code should be modular and maintainable, following React component best practices.

5. Methodology, Work Plan, and Team Roles

5.1 Methodology

The project follows a modular, iterative development approach. Each functional module — authentication, file upload, material browsing, and API integration — is developed, tested, and integrated independently before the final end-to-end integration and deployment phase. This approach ensures early detection of bugs and cleaner integration.

5.2 Project Timeline (12 Weeks)

Duration	Phase / Task
Week 1–2	Requirement Analysis, Database Design, UI/UX Wireframing
Week 3–4	Backend Setup: Node.js/Express server, JSON-based DB, file upload APIs
Week 5–6	Frontend Development: React components, Teacher Dashboard, role-based routing
Week 7–8	Student Dashboard, Material Download Feature, API Integration
Week 9–10	Testing (Unit + Integration), Bug Fixes, UI Refinement
Week 11–12	Deployment, Documentation, Final Review & Submission

Table 5.1: Project Timeline

5.3 Roles and Responsibilities

Student Name	Registration No.	Responsibilities
Harshal Modi	23BCON0673	Backend Development — Node.js/Express APIs, file upload logic, JSON database management
Siddharth Saxena	23BCON0636	Frontend Development — React components, routing, Teacher & Student Dashboards
Puneet	23BCON1987	Integration, Testing, Documentation, and Final Presentation

Table 5.2: Team Roles and Responsibilities

6. Conclusion and Future Scope

6.1 Conclusion

ExamBuddy represents a practical and impactful application of modern full-stack web development technologies in an academic setting. By providing a structured, role-based portal for sharing study materials, the platform solves a real and common problem faced by students and educators alike — the lack of a centralized, organized system for distributing course resources.

The project demonstrates the effective integration of a React frontend with an Express backend, the use of Multer for file handling, and JSON-based lightweight storage — making it an excellent learning exercise in building end-to-end web applications. The clean separation of roles ensures that the platform is intuitive and secure for both teachers and students.

6.2 Future Scope

- Integration with a full-scale database such as MongoDB or PostgreSQL for scalable and persistent storage.
- Authentication via JWT tokens or OAuth 2.0 for secure and verifiable role management.
- Search and filter functionality to allow students to find materials by subject, type, or topic keyword.
- Email notification system to alert students when new materials are uploaded by teachers.
- Mobile application development using React Native for seamless on-the-go access.

Declaration and Approval

We hereby declare that the project synopsis titled "ExamBuddy: A Full-Stack Web Platform for Sharing Study Materials Between Teachers and Students" has been prepared by us under the supervision of Mr. Divyansh Dewani, Assistant Professor from Department of Computer Science & Engineering, JECRC University, Jaipur. The content of this synopsis is original, and no part of this work has been copied from any other source or submitted elsewhere.

Students' Signatures:

Harshal Modi 23BCON0673	Siddharth Saxena 23BCON0636	Puneet 23BCON1987

Supervisor's Approval:

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